

Inverter (LT3462A) — Working Document

Digital Audio Player · Rev 5

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1. Part Identification

Field	Value
Manufacturer / MPN	Analog Devices — LT3462AES6
Package	6-lead TSOT-23 (S6)
Function	2.7 MHz current-mode inverting converter, integrated Schottky; +3V9 → −3.9 V (LT3093 input)

2. Design Requirements

Requirement	Value
Input	+3V9 from LTC3440, 3.827 – 3.981 V (min 3.74 V during load step)
Output	−3.9 V, regulated
Load	LT3093 only
Output current	≈ 27 mA quiescent; ≈ 98 mA peak (0.6 V RMS max); 125 mA LT3093 limit
Topology	Dual inductor inverting (Ćuk)
Switching frequency	2.7 MHz fixed (2.0 – 3.5 MHz), no Burst Mode
Enable	Follows +3V9 (SDREF internal pull-up); no separate shutdown control

Constraints: Low noise · Small footprint

3. Pin Configuration

Pin	Name	Connection
1	SW	L1, C1 (+)
2	GND	GND
3	FB	R1 / R2 midpoint, C4 to −3V9
4	SDREF	R2 to FB only — no GPIO (abs max 2.5 V), no other load
5	D	C1 (−), L2
6	VIN	+3V9, C_IN

4. External Components

Ref	Value	Rating / package	MPN	AL-PN
L1, L2	22 μ H $\pm$ 20%	Shielded, 300 mA rated, 400 mA I_{sat} , 810 m Ω max, 1210	Murata LQW32FT220M2HL	#1-090
C1 (flying)	1 μ F $\pm$ 10%	35 V X5R 0402	Murata GRM155R6YA105KE11D	#1-092
C_IN	10 μ F (6.76 μ F eff. at 3.9 V)	10 V X7R 0805	Murata GCM21BR71A106KE22K	#1-081
C_OUT	10 μ F (6.76 μ F eff. at 3.9 V)	10 V X7R 0805	Murata GCM21BR71A106KE22K	#1-081
R1	309 k Ω 0.1%	1/16 W 0402	YAGEO RE0402BRE07309KL	#1-091
R2	100 k Ω 0.1%	0603	YAGEO RT0603BRD07100KL	#1-066
C4	22 pF $\pm$ 5% (confirmed by load-step sim)	50 V C0G 0402	Murata GRM1555C1H220JA01D	#1-093

5. Equations and Results

Output voltage (R1 FB $\rightarrow$ V_OUT, R2 SDREF $\rightarrow$ FB)

$$V_{OUT} = -1.265 V \times \frac{R_1}{R_2} \Rightarrow R_1 = R_2 \times \frac{3.9}{1.265} = 308.3 k\Omega \rightarrow 309 k\Omega$$

$$V_{OUT} = -1.265 \times \frac{309}{100} = -3.909 V \quad (-3.816 \text{ to } -3.971 V \text{ over SDREF-FB } 1.235-1.285 V)$$

Divider current (SDREF can source 80 μ A)

$$I_{R2} = \frac{1.265 V}{100 k\Omega} = 12.7 \mu A$$

Duty cycle ($V_D = 0.8 V$ Schottky, $V_{CE} = 0.27 V$ switch)

$$D = \frac{|V_{OUT}| + V_D}{V_{IN} - V_{CE} + |V_{OUT}| + V_D} = \frac{4.709}{3.634 + 4.709} = 56.4\% \quad (57.6\% \text{ at } 3.74 V; \text{ max } 77\%)$$

Inductor ripple (each inductor, 22 μ H)

$$\Delta I_L = \frac{V_{IN} D}{f L} = \frac{3.904 \times 0.564}{2.7 MHz \times 22 \mu H} = 37 mA_{pp} \quad (50 mA_{pp} \text{ at } 2.0 MHz)$$

Inductor and switch current (98 mA peak output, $\eta = 0.65$ assumed)

$$I_{L1} = \frac{|V_{OUT}| I_{OUT}}{\eta V_{IN}} = \frac{3.909 \times 0.098}{0.65 \times 3.904} = 151 mA, \quad I_{L2} = I_{OUT} = 98 mA$$

$$I_{SW,pk} = I_{L1} + I_{L2} + \Delta I_L = 151 + 98 + 37 = 286 \text{ mA} \quad (299 \text{ mA at } 2.0 \text{ MHz})$$

Output current	Switch peak	vs 300 mA min / 420 mA typ limit
27 mA (quiescent)	$\approx 0.10 \text{ A}$	✓
69 mA (110 dB SPL peaks)	$\approx 0.21 \text{ A}$	✓
98 mA (full scale, 23.8 Ω)	$\approx 0.29 \text{ A}$	At minimum limit
125 mA (LT3093 limit)	$\approx 0.36 \text{ A}$	Inverter limits first

Inrush (undamped upper bound; Schottky surge limit 1.5 A)

$$I_P \leq V_{IN} \sqrt{\frac{C_1}{L_1}} = 3.9 \times \sqrt{\frac{1 \mu\text{F}}{22 \mu\text{H}}} = 0.83 \text{ A}$$

Step-input bound; the $\approx 6 \text{ ms}$ LTC3440 soft-start ramp keeps the real peak well below it.

Inductor copper loss (98 mA peak output, 810 m Ω max)

$$P_{DCR} = (I_{L1}^2 + I_{L2}^2) R_{DCR} = (0.151^2 + 0.098^2) \times 0.81 = 26 \text{ mW} \quad (I_{L1} = 151 \text{ mA vs } 300 \text{ mA rated})$$

Efficiency and currents (sim, 3.9 V in, nominal capacitor values, 0.81 Ω inductor DCR)

Item	98 mA load (39.9 Ω)	27 mA load (144.8 Ω)
V_OUT	-3.929 V	-3.938 V
Efficiency	67.7%	71.8%
Input power	572 mW	149 mW
I_L1 average (LTC3440 load)	141 mA	35 mA
Switch peak	0.282 A	0.100 A
Output ripple	0.52 mV pp	0.78 mV pp

Load step (sim, 27 $\leftrightarrow$ 98 mA, 1 μs edges, C4 stepped)

C4	Step up (27 $\rightarrow$ 98 mA)	Step down (98 $\rightarrow$ 27 mA)
18 pF	+28 mV (to -3.910 V)	-20 mV (to -3.958 V)
22 pF	+26 mV (to -3.912 V)	-18 mV (to -3.956 V)

Well damped, single overshoot. At the LT3093 output this is $\approx 10 \mu\text{V}$ (68 dB line-step rejection, LT3093 sim).

6. Important Values

Item	Value
Input range	2.5 – 16 V
Switch current limit	300 mA min / 420 mA typ (varies with duty cycle)
Switch V_CESAT	270 mV typ / 350 mV max at 250 mA
Schottky forward drop	800 mV typ / 1100 mV max at 250 mA
Maximum duty cycle	77% (LT3462A)
SDREF / reference	1.265 V (1.245 – 1.285 V), up to 80 μ A
FB bias current	15 nA typ / 50 nA max
Supply current	2.9 mA typ, not switching; < 10 μ A shutdown
Output	–3.909 V calc (–3.816 to –3.971 V); –3.93 V sim
Load regulation	–3.938 V at 27 mA $\rightarrow$ –3.929 V at 98 mA (9 mV, sim)
Load step 27 $\leftrightarrow$ 98 mA	+26 / –18 mV with C4 = 22 pF (sim)
Duty cycle	56.4% at 3.9 V in
Switch peak at 98 mA	0.282 A sim vs 300 mA min (6% margin)
Efficiency	67.7% at 98 mA, 71.8% at 27 mA (sim)
Inductor loading	L1 141 mA / L2 98 mA avg vs 300 mA rated (sim); peak $\approx$ 0.17 A vs 400 mA I_sat
θ_{JA}	150 $^{\circ}$ C/W
IC dissipation at 98 mA	$\approx$ 160 mW (185 mW loss – 24 mW inductor copper) $\rightarrow \approx$ 24 $^{\circ}$ C rise

7. Open Problems

#	Problem
1	Lead times: L1/L2 25 wk, R1 20 wk, C1 / C4 17 wk — order early
2	C1 effective capacitance at 7.8 V (0402 X5R derates heavily) — SimSurfing, then re-run efficiency and load step with that value
3	Soft-start C_S (SDREF to V_OUT) not required — place DNP footprint for bring-up?

8. Revision Log

Rev	Date	Change
0	2026-09-23	Initial issue: part selection, divider, duty cycle, switch current, inrush.
1	2026-09-23	L1/L2 LQW32FT220M2HL; C_IN 10 μ F #1-081; C1 4.7 μ F #1-071; SDREF to R2 only.
2	2026-09-23	L1/L2 assigned #1-090; I_sat 400 mA confirmed.
3	2026-09-23	C1 to 1 μ F $\geq$ 25 V. Efficiency and current sim added. Closed: switch peak, efficiency, dissipation.
4	2026-09-23	Load step sim added; C4 = 22 pF confirmed. Closed: C4 value, load step.
5	2026-09-23	Library parts assigned: R1 #1-091, C1 #1-092, C4 #1-093.